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An Online Reversed Bangla Sign Language Learning System

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***Abstract*—Hearing and speech impaired people around the world express their thoughts through sign language, which is incomprehensible to most people. As a result, interacting with them is quite challenging. Specially in Bangladesh, where people are unable to learn the Bangla sign language (BdSL) due to a lack of resources. We have proposed and implemented an online reversed Bangla sign language learning system using unsupervised rule-based method, which can be utilized to interpret Bangla natural language (speech and/or text) into BdSL. The system takes input of Bangla speech via a Bangla voice recognition module and then convert it into Bangla text. The system can also take Bangla text directly from the text editor module of the system. The system parse Bangla text using a designed set of rules from Bangla semantic analysis. After semantic analysis the system is trained based on unsupervised learning using the parsed Bangla text for each corresponding BdSL. We have used an efficient animation algorithm instead of video clips to display the BdSL for corresponding Bangla speech or text. We have considered only starting and ending frame of each gesture to make the animation. As a result, by using limited number of postures, the system can dynamically make any gestures based on the proposed animation algorithm. The system can animate sentences of various lengths and numerals ranging from 0 to 9 in BdSL. It can generate many phrases with a small quantity of data and has been trained and and tested with over 500 examples with lower computation cost rather than existing systems. The proposed system achieves 99.60% accuracy for speech/voice inputs and 100% accuracy for text inputs where as the average accuracy is 99.8% with the mean computational cost of 120 milliseconds per gesture animation.**

***Index Terms*—Bangla Sign Language, Animation, Rule Based Learning, Text Parsing.**

I. INTRODUCTION

Sign language (SL) is a language used by speech or hearing impaired people to communicate with general people [1]. . Sign language users are just as healthy as non-sign language users. They can understand, see and perceive everything, not simply talk. So the hearing or speech impaired people use sign language and symbolic instructions to communicate with non sign people in society. Depending on the language, sign language varies from region to region or country to country. There are currently over 200 sign languages around the world and approximately 7 million people with hearing and speech impairments[2] [3]. At present, there are one lakh 66 thousand and 397 registered hearing impaired individuals in Bangladesh. Of these, 1 lac 18 thousand 907 are speech impaired and 47 thousand 490 are hearing-impaired . Sign language plays a vital role in integrating this population into society and transforming them into a competent workforce. Language socialization relies on the ability of tangible communication to involve newbies in understanding and realizing both familiar and unfamiliar ways of thinking, feeling, and acting with others over the course of their lives [4]. However, it is difficult for hearing or speech impaired people to socialize with the mainstream due to the language barrier. Therefore, we have introduced an online reversed Bangla sign language learning system using unsupervised Rule Based method to enable the normal people to overcome this language barrier and communicate with the hearing and speech impaired people.

Both voicing and text input are supported by the proposed approach. The system parses the data using set of designed Bangla semantic rules and then animated the relevant sign language by implementing an efficient animation algorithm to make dynamic animated gestures instead of video clips which makes the system faster and more accurate. By using our developed API, the proposed system is favored as it can be easily used in android system. The rest of the paper is organized as follows: Section II discusses related works.Proposed method is discussed in Section III. Result analysis is presented in Section IV. Finally, paper is concluded in Section V.

II. RELATED WORK

All over the world, researchers have created many systems in different languages for sign people. However, no significant work especially mobile application has been done in our country to enable sign and non-sign people communication[5][6]. However, there have been recent breakthroughs in the field and the research is only growing. Shahriar et al. Developed a system. They first took voice input in the proposed system [7]. They used Google Translator to convert input voice to text. They have used fixed speech and external noise affects the accuracy of their system. Md. Tazimul haque et al. Developed a system. Their proposed system first inputs Bangla sentences or words. Then BdSL processes the input through grammar [8]. Bengali sign language has been selected from the database and converted into sign language. They have used limited words for Tests and have not used dynamic process. Their application shows 95% of test results.

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Md Masud et al [9] developed a system which made animation by selecting Bangla sign language from the database for each text input. They have used a small number of fixed animation videos for Bangla Sign having 80% accuracy.
Ishteaque Alam et al. [10] developed a system which works on pre-determined grammatical rules with voice as input and displays animated signs achieving 99% accuracy.
S. M. Halwani [11] has proposed an Arabic sign language translation system achieving accuracy of 95.62%. Promila Haque et al. [12] proposed a system consisting two-handed Bangla Sign Language Recognition, the three-stage structure of the system, training, and classification. It can recognize 26 sign cues . They used 104 images for testing and achieved 98.11% accuracy.Sakshi Lahoti et al.[13] proposed a smartphone platform for ASL (American Sign Language) recognition by appling Cany Edge detection. There are 36 symbols in their database for ASL. The accuracy achieved by the system has become 89.54%.

Zakariya et al. [14] used smartphone as a platform in their proposed system which is able to classify 10 Arabic sign languages with satisfactory performance.

III. PROPOSED METHODOLOGY

The system will take input from users and process the text and animate the interpretation of the input in Bangla Sign Language to the Bangla Sign Data from the cloud storage through API. We have implemented BdSL recognition through an Android app. We have used the android studio software for smartphone applications. Figure 1 shows the architecture of the system.

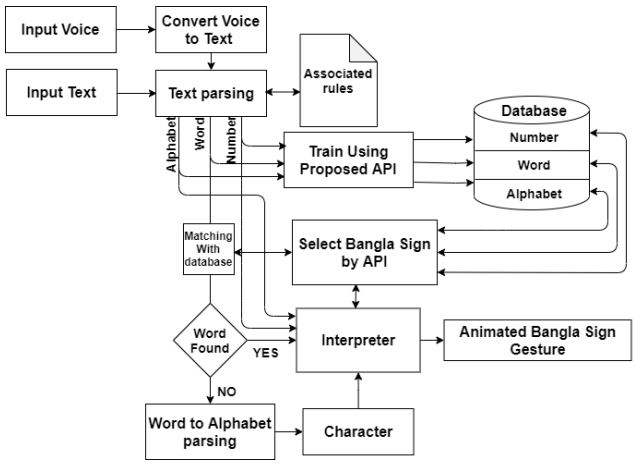


Figure 1: Proposed system architecture

A. Input and Text Process

The system can take input in both ways: voice and text. Users gives the Bangla voice or text as input through smart-phone applications. After manually taking input from the user, the entire text is divided into Word using the proposed algorithm 1. That is shown in Figure 3. And finally, they are stored inside an array. Which is shown in Table I.

Algorithm 1: Algorithm for text parsing

```
Initialization: text=recognitionText,
  length=text.length, loop variable i, array words[100];
Function: textParsing()
for i=0 to length do
    if text[ i ] == space then
        words=text[ i ]trim;
    end
end
End Function
```

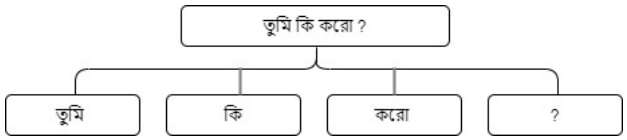


Figure 2: Manual text processing system

1) Voice Recognition: Google's default speech recognition system is used to receive the voice input and then convert it into Bangla text. After voice recognition, the text is divided into words using the proposed algorithm 2, as like as text input processing. But there are some differences as we have used Google's default speech recognition system. Since question mark can not be given as input through voice we have we have added question marks manually after converting the voice into text as shown in the Figure 4, 5 and Table II.

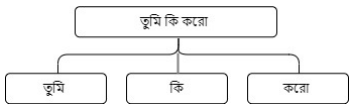


Figure 3: Recognized text processing system

B. Selection Process For Bangla Sign Language

In this part we have highlighted the process of Bangla Sign Language Selection in the proposed system. Algorithm 3 is the main algorithm for selecting postures tor making dynamic gesture animation corresponding Bangla text. It offers algorithm 4, 5, and 6 for selection of postures to make animated gestures for words, characters and numerals respectively. Algorithm 3 selects the sign data from the cloud database via API. And Algorithm 4 compares the sign data with the data we give input. And select the sign languages. Then animate the selected sign languages.

Table I: Input and array position to keep record after text parsing

Array Index	Words
0	তুমি
1	কি
2	করো
3	?

Algorithm 2: Algorithm for voice to text processing and parsing

```
Initialization: text=recognitionText,
  length=text.length, loop variable i, array
  words[100],flag=0;
Function: checkQuestionableAndParsingText()
for i=0 to length do
    if text[ i ] == space then
        words=text[ i ].trim ;
        if words [ i ] == questionable then
            flag++;
        end
    end
end
if flag greater-than 0 then
    words[ length + 1 ] = ? ;
end
End Function
```

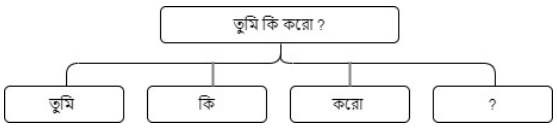


Figure 4: Recognized text processing system having ”?” marks

C. Training of The Proposed System

We have trained the system through different types of data sets using algorithm 7, 8 and 9 for Bangla word, character numbers respectively. Figure 6, 7and 8 represent some graphical representations of the data set. We worked with 51 Bangla Characters that are represented by only 36 BdSL but showed 9 for example due to page constraints. And worked from 0 to 9 but showed 9 as an example.

IV. PERFORMANCE EVALUATION

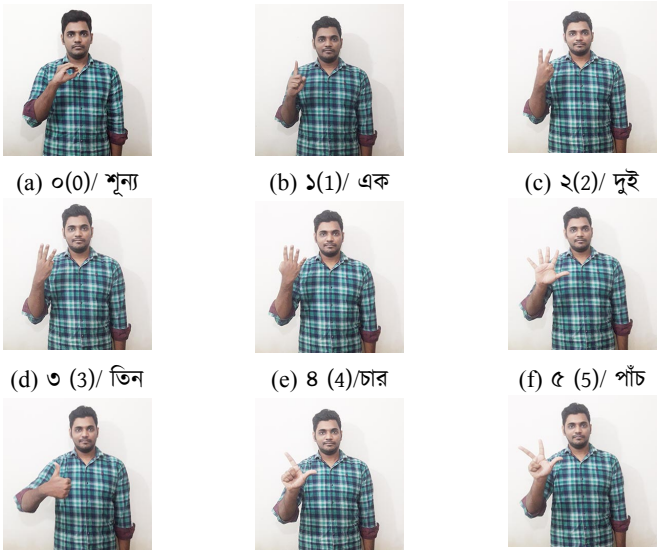
The Experiment is performed on the dataset containing over 500 Bangla word gestures, 36 character signs and 0-9 number signs. In BdSL, one sign can be used to express multiple expression. We have considered 0-9 Bangla numbers, 36 Bangla characteters 500 Bangla words and diverse 100 Bangla sentences including interrogative sentences (গাড়ি কোথায়?, তুমি কি করো?,...) etc. where each class contains 100 samples of data to evaluate the system. We have also measured the performance of the system based on the input from both voice

Table II: Input and array position sentence parsing having ”?” marks

Array Index	Words
0	তুমি
1	কি
2	করো
3	?

Algorithm 3: Main algorithm for postures selection to make animated gestures

```
Initialization: n =
  input.length,singleword,str=input[0],input[],ch[] ;
Function: testingAndTrainingData()
if input.length equal 1 then
    for i=0 to i<input[0].length() increment do
        ch[i] = str.charAt(i);
    end
    if ch.length equal 1 then
        for i=0 to i<=9 increment do
            if ch[0] equal i then
                numberTraining(i);
                String signdata = numberSelection(i);
                Output sign;
                break;
            else
                charTraining(ch[0]);
                String signdata = charSelection(ch[0]);
                Output sign;
                break;
            end
        end
    else
        wordTraining();
        String signdata = wordSelection();
        Output sign;
    end
else
    wordTraining();
    String signdata = wordSelection();
    Output sign;
end
End Function
```



(a) ০(0)/ শূন্য (b) ১(1)/ এক (c) ২(2)/ দুই

(d) ৩ (3)/ তিন (e) ৪ (4)/চার (f) ৫ (5)/ পাঁচ

(g) ৬ (6)/ ছয় (h) ৭ (7)/ সাত (i) ৮(2)/ আট

Figure 5: Example Dataset of Bangla number signs

Algorithm 4: Algorithm for Bangla Word selection

```
Function: wordSelection()
if words[n]== ?) then
    return, n = n - 1;
end
while i != n do
    singleword = n [ i ];
    Connect to word get api; for i = 0 to
     wordDBlength do
        if singleword == worddatalist [ j ] then
            return, worddatalist [ j ];
        else
            for ) do  i = 0 to i<singleword.length(
                char allChar[i] = singleword.charAt(i);
            end
            Whilej != allChar.length() singlechar =
             allChar[ j ];
            Connect to character get api; for k = 0 to
             charDBlength do
                if singlechar == charSignDataList[ k ]
                 then
                    return, charSignDataList[ k ];
                end
            end
        end
    end
end
End Function
```

Algorithm 5: Algorithm for Bangla character selection

```
Function: charSelection(char c)
Connect to character get api;
int charDataLength = charsigndataList.length;
for k = 0 to charDataLength do
    if c == charsigndataList[ k ] then
        return, charsigndataList[ k ];
    end
end
End Function
```

Algorithm 6: Algorithm for Bangla number selection

```
Function: numberSelection(int num)
Connect to number get api;
int numberDataLength = numbersigndataList.length;
for k = 0 to numberDataLength do
    if num == numbersigndataList[ k ] then
        return, numbersigndataList[ k ];
    end
end
End Function
```

Algorithm 7: Training algorithm for Bangla Word learning

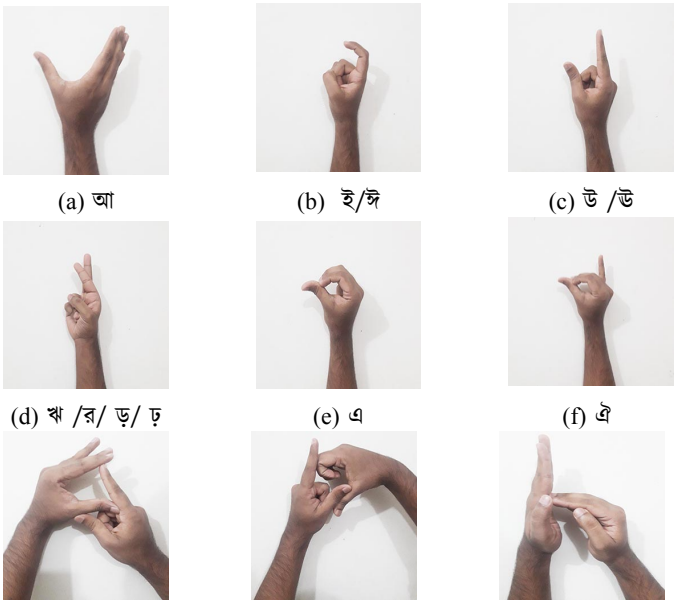
```
Function: wordTraining()
if words[n]== ?) then
    return, n = n - 1;
end
while i != n do
    singleword = n [ i ];
    Connect to word post api;
    setdata if not exit signkey = singleword;
    signimage = pick image form storage;
end
End Function
```

Algorithm 8: Training algoritm for Bangla character learning

```
Function: charTraining(char c)
Connect to char post api;
setdata if not exit, signkey = c;
signimage = pick image form storage;
End Function
```

Algorithm 9: Training algorithm for Bangla number learning

```
Function: numberTraining(int num)
Connect to number post api;
setdata if not exit, signkey = num;
signimage = pick image form storage;
End Function
```



(a) আ (b) ই/ঈ (c) উ /ঊ

(d) ঋ /র/ ড়/ ঢ় (e) এ (f) ঐ

(g) ক (h) খ/ক্ষ (i) গ

Figure 6: Example Dataset of Bangla character signs where only 9 signs are shown among 36 signs due to page limit

Figure 7: Example Dataset of Bangla Word signs

and text modules for Bangla number, words and sentences. For Bangla chracters, we have only considered the text inputs. The accuracy of the system has been determined by 1. TP (True Positive), TN (True Negative), FP (False Positive), and FN (False Negative) [1] have been calculated from several confusion matrices which are not shown in here due to large size.

$$Accuracy = \frac{TP \ + \ TN}{TP \ + \ TN \ + \ FP \ + \ FN} \times 100 \qquad (1)$$

We have also measured precision, recall and F1-Score of the system by the followings.

$$Precision = \frac{TP}{TP \ + \ FP} \times 100 \qquad (2)$$

$$Recall = \frac{TP}{TP + FN} \times 100 \qquad (3)$$

$$F1 - score = \frac{2 \times precision \times recall}{precision + recall} \qquad (4)$$

Table III shows the accuracy of Bangla numbers where inputs are taken from voice. For the Bangla alphabet and Numbers interpretation, the system achieves 100% accuracy for each test cases if the inputs are taken from manual text input system module, That's why we have ignored to present results in tables for text input. Table IV and V show the accuracy of ten sample data set of Bangla words and sentences respectively where inputs are taken from voice. We have considered only 10 samples among 100 testing data where each class contains 100 test data.

We have also measured the accuracy of the Bangla spelling and word through which are shown in Table VI and Table VII

Table III: Experimental result for example set of Bangla Number taken from voice input

Data List	Total Test	TP	FP	FN	TN	Accuracy (%)
০(0) / শূন্য	100	98	2	0	0	98
১(1) / এক	100	99	1	0	0	99
২(2) / দুই	100	99	1	0	0	99
৩(3) / তিন	100	98	2	0	0	98
৪(4) / চার	100	98	2	0	0	98
৫(5) / পাঁচ	100	97	3	0	0	97
৬(6) / ছয়	100	97	3	0	0	97
৭(7) / সাত	100	98	2	0	0	98
৮(8) / আট	100	98	2	0	0	98
৯(9) / নয়	100	96	4	0	0	96

Table IV: Experimental result for example set of Bangla words taken from voice input

Data List	Total Test	TP	FP	FN	TN	Spelling Accuracy (%)
কাজ	100	97	3	0	0	97
কোথায়	100	97	3	0	0	97
গাড়ি	100	96	4	0	0	96
ঘর	100	98	2	0	0	98
ঘুমাবো	100	97	3	0	0	97
ভাই	100	97	3	0	0	97
বাংলাদেশ	100	99	1	0	0	99
সুন্দর	100	98	2	0	0	98
সূর্য	100	97	3	0	0	97
অন্ধকার	100	97	3	0	0	97

Table V: Experimental result for example set of Bangla sentence taken from voice input

Data List	Total Test	TP	FP	FN	TN	Word Accuracy (%)
আমি ভাত খাই	100	98	2	0	0	98
তুমি কি করো?	100	98	2	0	0	98
কোথায় যাবে?	100	97	3	0	0	97
বেড়াতে যাবো	100	97	3	0	0	97
আপনাকে ধন্যবাদ	100	97	3	0	0	97
আমি আত্মবিশ্বাসী	100	97	3	0	0	97
আমার সোনার বাংলা	100	97	3	0	0	97
আজ বৃষ্টি হবে না	100	97	3	0	0	97
স্কুলে যাও	100	97	3	0	0	97
পরীক্ষা হবেনা	100	97	3	0	0	97

shows the accuracy of the system based on types of input. Table VIII shows the computational cost by comparing the state-of-the-art method with our proposed work. It shows that, the system performed admirably with the mean accuracy of 99.8% and outperforms the study [9]. The proposed system is about 4 times faster than the system [9] with keeping higher accuracy due to use of efficient animation algorithm instead of using video clips of animated gestures.

Table VI: Experimental outcome for Bangla word spelling in (%)

Form of Inputs	Accuracy	Precision	Recall	F1- Score
Spelling	99.8	100	99.7	99.8
Word	100	100	100	100

Table VII: Performance evaluation of the system based on inputs from speech/voice and text in (%).

Form of Inputs	Accuracy	Precision	Recall	F1- Score	Average Accuracy
Voice	99.6	99.7	99.8	99.7	
Text	100	100	100	100	99.8

Table VIII: Comparison analysis of the performance of the proposed system with state-of-the-art method based on mean accuracy rate and Computational cost or delay time in millisecond (ms) per gesture animation

Performance Parameters	Proposed System	State-of-the-art method [9]
Delay Time (ms)	120	500
Mean Accuracy (%)	99.8	94.71

V. CONCLUSION

This paper has proposed and implemented an online reversed Bangla sign language (BdSL) learning system to make an effective communication between sign and no-sign people. Through smart phones, the system can translate Bangla speech and text into animated BdSL. The system can generate a large number of dynamic animated gestures corresponding any Bangla sentences from a small quantity of training data (36 character signs, 0-9 number signs and 500 word gestures). It can also deal with interrogative sentences given via voice. The system's trained database is hosted on the cloud. We've also created an API for accessing the database. Our method reduces the distance of communication between signs and non-sign peoples, or helps them to learn Bangla sign language in the revers way. Regardless of the fact that it performs admirably with the accuracy of 99.8%, there is still potential for advancement in animation module to make more faster. In the future, the dataset can be improved and added 3D model dataset to recognize the gestures for corresponding Bangla words used in everyday life and make them more understandable.

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